A historical Stage-2 synthesis draft for TECT (Math60-A..E old packaging)

(NOT a current canonical Stage-2 closure paper; the current canonical Stage-2 composite tier is T3 per Math307 / Math310 with promotion gated by $(C_1 \wedge C_2 \wedge C_3) \wedge (\textbf{F-GAP4} = \textbf{PASS})$. External use FORBIDDEN at the present canonical tier per Math314-AddD audit.)

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This document is preserved as a historical Stage-2 synthesis draft (Math60-A..E old packaging) for the Topological Energy Condensate Theory (TECT). External use is FORBIDDEN at the present canonical tier per the Math314-AddD audit, because the original abstract / introduction wording ("Stage-2 global closure theorem" / "quantum completion" / "structural integrity established") is incompatible with the current canonical bookkeeping:

$$\text{GAP-2} = T6,\ \text{GAP-3} = T6,\ \text{GAP-1} = T4,\ \text{GAP-4} = T3, \quad \text{Stage-2 composite} = \min = T3.$$

Stage-2 promotion requires the joint event $(C_1 \wedge C_2 \wedge C_3) \wedge (\text{F-GAP4} = \text{PASS})$ per Math307 / Math310 / Math311 latest synthesis; none of these gates have arrived at the present date. The original Math60-A..E five-sub-theorem packaging predates the GAP-1/2/3/4 + verdict-shell decomposition that Math270–313 introduced; this draft is therefore a chronological-record artefact, not a current canonical claim. Refer to the canonical `Docs/status/TOE-FACT-SHEET.md` and to Paper-12 successor drafts (when available) for the current Stage-2 status.

I. INTRODUCTION — HISTORICAL RECORD FRAMING

Historical scope. This document records the *Math60-A..E (2026-04 era) Stage-2 synthesis packaging* as it was drafted prior to the Math270–Math313 verdict-shell decomposition. At that time, "Stage-2 closure" was framed as the conjunction of five sub-theorems M60-A, . . . , M60-E. The packaging was overtaken in late April / early May 2026 by the GAP-1/2/3/4 + verdict-shell architecture introduced in Math270 onwards, in which Stage-2 promotion is gated by the joint event $(C_1 \wedge C_2 \wedge C_3) \wedge (\text{F-GAP4} = \text{PASS})$. The current canonical Stage-2 composite tier per Math307 / Math310 is **T3 proof sketch**, with the four-GAP sub-tiers

$$\text{GAP-2} = T6, \quad \text{GAP-3} = T6, \quad \text{GAP-1} = T4, \quad \text{GAP-4} = T3.$$

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Because the present paper's body uses the older Math60 packaging, it *cannot* be promoted to or used as a current canonical Stage-2 closure paper.

Reading guide. The remainder of this document reproduces the 2026-04 era Math60 packaging verbatim, with each section now introduced by an explicit "Historical record" header noting the gap between the older claim and the current canonical bookkeeping. A reader interested in the current Stage-2 status should consult `Docs/status/TOE-FACT-SHEET.md` and the canonical Math307 / Math310 / Math311 verdict-shell archives, not the body of this paper.

II. HISTORICAL SETTING: THE 11-PILLAR FRAMEWORK (MATH60-A..E PACKAGING)

Historical record. The 2026-04 era Math60 packaging listed eleven foundational closure conditions, reproduced verbatim below. The list has *not* been re-audited against the current Math270–Math313 verdict-shell decomposition; it should be read as a chronological record, not as a current canonical roster.

A. The 11-pillar framework (historical Math60 packaging)

TECT posits 11 foundational closure conditions:

1. BCC ground-state uniqueness (mass gap m^*)
2. Kinematic Lorentz invariance at the BCC fixed point
3. Emergent gravity (spin-2 graviton at one loop)
4. SO(10) gauge-group emergence
5. Chirality from index-theoretic protection
6. Generations and SM embedding
7. Per-generation quantum consistency (Ward + Witten + mod-2)
8. Lorentz invariance (interval certificate)
9. Equivalence principle from maximum-proper-distance principle
10. Planck constant $\hbar$ origin (phase transition + matter side)

11. Cosmological constant Λ four-sector cancellation

In the Math60 packaging, "Stage-2 closure" was framed as a *compatibility* statement on the 11-pillar system, not as independent verification of each pillar.

B. Historical Math60 composite (NOT a current theorem)

Remark 1 (The following theorem statement is preserved as a historical artefact and is NOT a current canonical claim). The text below reproduces the Math60-era Stage-2 composite statement verbatim. It was correct as a packaging proposal at the 2026-04 canonical tier but has been *superseded* by the GAP-1/2/3/4 verdict-shell decomposition of Math307 / Math310 / Math311. Promoting this statement to a current canonical claim is forbidden by the Math314-AddD audit and by the canonical-source hierarchy (the internal canonical-source hierarchy). External use is FORBIDDEN at the present canonical tier.

a. Historical Math60 composite (verbatim, 2026-04 packaging, NOT current canonical). The Math60-era proposal defined the TECT global qualification predicate as the conjunction

$$S_2^{\text{Math60-era}} := \text{M60-A} \wedge \text{M60-B} \wedge \text{M60-C} \wedge \text{M60-D} \wedge \text{M60-E},$$

where:

1. **M60-A (Meta-consistency)**: The 11-pillar decomposition admits a pairwise-commutativity diagram showing no logical cycles or circularity under the BCC condensate assumption [**?**]. Status: T6 PROVED CONDITIONAL.

2. **M60-B (Parameter compression)**: The TECT effective axiom count N_{axiom} reduces from 3 independent postulates to $2 + 1_{\text{cosmological}}$ under the Meta-consistency diagram. Compression ratio $\rho_{\text{compression}} = 4.75\times$ quantifies the redundancy elimination. Status: T6 PROVED CONDITIONAL.

3. **M60-C (Quantum observables)**: Three canonical quantum observables (QO1, QO2, QO3) are uniquely determined by the classical condensate structure, with closed-form extraction rules (Casimir operator, noise spectrum, form factor). Status: T4 STRONG EVIDENCE.

4. **M60-D (Observable-map injectivity)**: The quantum-to-classical map from 5 quantum observables to a 9-dimensional classical observable vector is globally injective. The Jacobian matrix is 5×9 with full column rank (rank $= 5$). Status: T6 PROVED CONDITIONAL.

5. **M60-E (Falsifiability)**: Four independent falsification gates (F1, F2, F3, F4) bind quantum predictions to classical cosmological observations. Status: T3 PROOF SKETCH (gates marked, closure pending).

The Math60-era composite is a T3 PROOF SKETCH. The 2026-04 packaging then conjectured that, under (H1) Pillars 1–9 being individually closed or STRONG-EVIDENCE, and (H2) the five sub-theorems being independently verified, TECT would be "mathematically self-consistent and falsifiable".

b. Status of the Math60 composite under the current canonical bookkeeping. Under the current Math270–Math313 verdict-shell architecture, the Math60 composite is replaced by the GAP-1/2/3/4 conjunction. The correspondences are

$$\text{M60-A, B} \longleftrightarrow \text{GAP-2,3 (T6 each)}, \quad \text{M60-C, D} \longleftrightarrow \text{GAP-1 (T4)}, \quad \text{M60-E} \longleftrightarrow \text{GAP-4 (T3)}.$$

The current canonical Stage-2 composite is therefore $\min(T6, T6, T4, T3) = T3$, matching the Math60-era $T3$ verdict but for a structurally different reason (the limiting tier is now driven by GAP-4 falsifiability package readiness and by the GAP-1 verdict-shell consumption). The Math60-era "H2: five sub-theorems independently verified" clause is replaced by the verdict-shell joint event $(C_1 \wedge C_2 \wedge C_3) \wedge (\text{F-GAP4} = \text{PASS})$.

III. HISTORICAL RECORD: MATH60 SUB-THEOREM PROOF-SKETCHES (VERBATIM, NOT CURRENT CANONICAL)

Historical record. The five proof-sketches below reproduce the Math60-era arguments. They are NOT re-audited against the current canonical Math270–Math313 archive; the reader should consult the canonical Math notes for the current state of each claim.

a. Step 1 (Meta-consistency diagram, M60-A historical). [**?**] constructs an explicit pairwise-commutativity diagram among the 11 pillars, showing that derivations taken in different orders reach the same conclusion. No logical cycles are found. The diagram is closed at T6 (rigorous proof with explicit hypothesis set).

b. Step 2 (Parameter compression, M60-B historical). [**?**] analyzes the axiom dependencies and identifies that A2 (the ultra-high-energy initial state) is partially derivable from A0 (BCC Brazovskii dynamics) + A1 (TDGL kinetics) given the cosmological cooling curve $T(t)$. Thus, the irreducible axiom count is 2 core + 1 cosmological boundary condition. Compression ratio = 4.75×. Status: T6.

c. Step 3 (Quantum observables closure, M60-C historical). [? ? ? ?] derives three canonical quantum observables (QO1 = Casimir, QO2 = noise spectrum, QO3 = form factor) with closed-form extraction rules from the classical condensate solution. QO1 and QO2 are analytically closed at T4 level. QO3 includes a numerical-anchor term. Status: T4 STRONG EVIDENCE.

d. Step 4 (Observable-map injectivity, M60-D historical). [? ? ?] proves that the map from 5 quantum observables to a 9-dimensional vector of classical cosmological observables is globally injective. The Jacobian is 5×9 with analytically verified column rank = 5. Status: T6 PROVED CONDITIONAL.

e. Step 5 (Falsifiability gates, M60-E historical). [?] defines four falsification gates F1–F4 that bind quantum predictions to classical observables. Each gate has a pre-registered closure deadline and acceptance criterion. The gates were marked T3 (proof sketch level) at the 2026-04 canonical tier; closure awaits empirical testing or theoretical verification. Under the current canonical bookkeeping these gates are subsumed by the GAP-4 falsifiability package, whose composite tier is T3.

IV. HISTORICAL SUB-THEOREM STATUS SUMMARY (2026-04 ERA; NOT CURRENT CANONICAL)

Historical record. The table below records the 2026-04 era Math60 sub-theorem tiers verbatim. The current canonical tiers are recorded in `Docs/status/TOE-FACT-SHEET.md` and the canonical Math notes; they may differ.

Sub-theorem	Title	Historical (2026-04)	Evidence (historical)
M60-A	Meta-consistency	T6	Explicit pairwise diagram
M60-B	Parameter compression	T6	Axiom dependence analysis
M60-C	Quantum observables	T4	Analytical + numerical anchor
M60-D	Observable-map injectivity	T6	Jacobian rank verification
M60-E	Falsifiability gates	T3	Gates marked; closure pending
	Historical Math60-era composite $S_2^{\text{Math60-era}}$: T3 PROOF SKETCH		

TABLE I. Historical Math60-era sub-theorem status summary (2026-04 packaging). The corresponding current canonical GAP-1/2/3/4 composite is $\min(T6, T6, T4, T3) = T3$ per Math307 / Math310.

A. Historical open gates and deadlines (2026-04 era)

- F-GAP2 (M60-E §3.2): Quantum–classical correspondence gate (historical deadline: 2026-05-22).
- F-GAP3 (M60-E §3.3): Cosmological-observable consistency (historical deadline: 2026-05-29).
- F-GAP4 (M60-E §3.4): Falsifiability package closure (historical deadline: 2026-06-05).

For the current canonical state of these gates, consult `Docs/status/OPEN-QUESTIONS.md` and the canonical Math3xx verdict-shell archives.

V. DISCUSSION — HISTORICAL ARTEFACT, NOT A CURRENT CANONICAL STAGE-2 VERDICT

The Math60-era Stage-2 packaging recorded above represented the 2026-04 best-effort synthesis of the TECT internal-architecture audits. It is preserved here as a chronological artefact; it is *not* a current canonical Stage-2 verdict and *must not* be cited as such. The current canonical Stage-2 composite tier is **T3 proof sketch**, with the limiting tier driven by GAP-1 (T4) verdict-shell consumption (Math299 / Math312) and by GAP-4 (T3) falsifiability-package readiness (Math311 / Math313). Promotion to T6 PROVED CONDITIONAL requires the joint event $(C_1 \wedge C_2 \wedge C_3) \wedge$ (F-GAP4 = PASS).

External use of this paper is FORBIDDEN per the Math314-AddD audit. A future canonical Stage-2 closure paper will replace the present historical record once the joint-event verdict is consumed.

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[] J. Lee, TECT-Math60-A: pairwise-commutativity diagram, TECT internal note (2026).

[] J. Lee, TECT-Math60-B: axiom dependency analysis, TECT internal note (2026).

[] J. Lee, TECT-Math60-C: three derivations, TECT internal note (2026).

[] J. Lee, TECT-Math60-C-Addendum-A, TECT internal note (2026).

[] J. Lee, TECT-Math60-C-Addendum-B, TECT internal note (2026).

[] J. Lee, TECT-Math60-C-Addendum-C, TECT internal note (2026).

[] J. Lee, TECT-Math60-D: map theorem, TECT internal note (2026).

[] J. Lee, TECT-Math60-D-Addendum-B, TECT internal note (2026).

[] J. Lee, TECT-Math60-D-Addendum-C, TECT internal note (2026).

[] J. Lee, TECT-Math60-E: F1–F4 falsification gates, TECT internal note (2026).